# Awanish Kumar Rai

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## SKILLS

- **Languages**: C, C++, Python, Java, JavaScript
- **Frameworks**: React, Express
- **Tools**: Git, GitHub, Google Colab, Linux, Windows, MS Excel
- **Web Technologies:** REST APIs, JSON, Node.js
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Specialized Skills:** Data Structures & Algorithms, Operating Systems, Networking
- **Soft Skills**: Problem-Solving, Adaptability, Time Management, Team Collaborations

## PROJECTS

**OS based Room Booking System | Link** Nov'2025 – Dec'2025

- Developed a concurrent backend system to handle simultaneous room booking requests with fair resource allocation and consistency guarantees.
- Implemented FCFS scheduling using semaphores and multithreading to prevent race conditions and ensure thread-safe access to shared resources.
- Designed REST APIs for booking, availability tracking, and queue visualization, enabling real-time monitoring of system state.
- Tech: Python, Django REST Framework, Threading, Semaphores

**My-Bookings | Link** Aug'2025-Sept'2025

- Built a production-grade booking system using MERN + Python scheduler, supporting horizontal scaling and real-time updates.
- Ensured zero double-bookings with MongoDB ACID transactions and atomic rescheduling logic.
- Achieved sub-100ms availability queries using Redis caching with smart invalidation.
- Enabled secure real-time updates via Socket.io + JWT, containerized with Docker for scalable deployment.
- Tech: React, Node.js, Express, MongoDB, Redis, Socket.io, Python, Docker

## TRAINING

Sumer PEP Training-Programing Pathshala & Lovely Professional University | Certificate

- Completed intensive summer training focused on problem-solving, algorithms, and core programming fundamentals.
- Built AlgoPhoto, an algorithmic photo processing system using custom logic and dynamic programming, avoiding heavy image-processing libraries
- Implemented pixel-level image transformations with emphasis on time and space complexity optimization.

## CERTIFICATES

- Computer Communications- University of Colorado Systems – Coursera | Certificate Sept'2025
- TCP/IP and Advanced Topics - University of Colorado Systems – Coursera | Certificate Sept'2025
- The Bits and Bytes of Computer Networking - Google – Coursera | Certificate Sept'2025
- Introduction to Hardware and Operating Systems - IBM– Coursera | Certificate Sept'2025

## ACHIEVEMENTS

- Achieved LeetCode rating of 1635 Dec'2025
- Achieved CodeChef rating of 1408 (2★) Nov'2025
- Achieved Codeforces rating of 1285(Pupil) Oct'2025

## EDUCATION

- **Lovely Professional University** Punjab, India
  Bachelor of Technology - Computer Science and Engineering ;
  CGPA: 8.29 Since Aug'2023
- **Sant Longowal Institute Of Engineering and Technology** Punjab, India
  Diploma in Chemical Technology; CGPA: 9.15 Aug'2020 – Jun'2023
- **Cambridge Senior Secondary School** Buxar, Bihar
  Matriculation; Percentage: 85.8% Apr'2019 – Mar'2020